MANISH KUMAR

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LinkedIn | Portfolio: https://mdigitechnology.github.io | GitHub

PROFESSIONAL SUMMARY

AI Engineer with 6+ years of experience building **production-grade computer vision and secure ML systems** for defense (DRDO) and infrastructure (NHAI/MoRTH). Skilled in **adversarial robustness (IBM ART), model explainability (Captum)**, and real-time edge deployment on NVIDIA Jetson. Currently focused on **Generative AI (LLMs, RAG, prompt engineering)** with hands-on project experience.

SKILLS

Generative AI & LLMs:
LLMs, RAG, Prompt Engineering, Conversational AI, NLP

ML/DL & Computer Vision:
PyTorch, TensorFlow, CNNs, Deep Learning, OpenCV, YOLOv4

Secure & Explainable AI:
IBM ART (Model Evasion, Extraction, Poisioning PGD, CW, FAB, Adversarial Robustness, Captum

Deployment & Backend:
NVIDIA GPUs Jetson (Xavier NX, Nano, Orin NX), Docker, Linux, Git, REST APIs, FastAPI, Django

Programming & Data:
Python, NumPy, Pandas

PROFESSIONAL EXPERIENCE

AI Engineer

GVR Technolabs Pvt. Ltd. *(DRDO – Ministry of Defence Projects)*
New Delhi | Oct 2024 – Present

- Developed secure **CNN-based vision models** with adversarial defenses (PGD, CW, FAB), improving robustness by ~30–40%
- Built **explainability pipelines (Captum)** and automated adversarial testing workflows
- Worked on **model optimization and validation for secure deployment scenarios**

R&D Engineer – AI/ML

VaaaN Infra Pvt. Ltd. *(NHAI/MoRTH Smart Highway Projects)*
Faridabad | Aug 2022 – Oct 2024

- Enhanced **Vehicle Incident Detection System (VIDS)** using CNN-based object detection for events such as wrong direction, vehicle stoppage, and over-speeding
- Contributed to **Automatic Traffic Counter & Classifier (ATCC)** for vehicle detection and classification (car, bus, truck, bike)
- Applied **YOLO-based models** for real-time video analytics
- Developed **web-based monitoring system** using React.js, Django, and REST APIs
- Deployed and tested models on **NVIDIA GPUs, Jetson devices (Nano, Xavier NX, Orin NX) and server environments**
- Worked with **camera-based traffic data** for real-time detection and monitoring systems

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Earlier Experience

IT Administrator & System Engineer (2018 – 2022)

- Worked on Linux systems, networking, and infrastructure support

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PROJECTS

AI ChatBot (LLM-based)

- Built conversational AI system using **LLMs** for context-aware responses and basic task automation.

Adversarial Robustness Framework

- Developed testing framework using **IBM ART and Captum** to evaluate and improve CNN model robustness.

VIDS & ATCC Systems

- Implemented **real-time traffic monitoring and incident detection** using computer vision on edge devices.

EDUCATION

Master of Computer Applications (MCA)
Swami Vivekanand Subharti University — 2022

B.Sc. Information Technology
Maharishi University — 2018

CERTIFICATIONS

- Neural Networks and Deep Learning – DeepLearning.AI
- Python for Data Science & AI — IBM
- AWS Certified DevOps Engineer – Professional
- Linux Administration Bootcamp